

Pregame Pitcher Strikeout Projection — Project Summary

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Companion to the full technical manuscript in `docs/paper/manuscript.md`.

Goal

Estimate a starter’s strikeout rate before first pitch, project how many batters that starter will face, and convert the pair into expected strikeouts and $P(K \geq L)$ for common prop lines L —using only information available before first pitch:

$$k_{\text{rate}} \times \text{TBF} \rightarrow E[K] \rightarrow P(K \geq L)$$

Models and headline metrics

Component	Model	Role
Strikeout rate	Unweighted LightGBM (180 features)	Frozen production rate model
Batters faced (TBF)	Ridge (thin bullpen, 24 features)	Projected exposure
Counts / lines	Binomial / Poisson on projected TBF	Expected K and $P(K \geq L)$

Frozen rate model (2023–2024 chronological test): MAE / RMSE / $R^2 \approx$ **0.0787 / 0.0987 / 0.147**. Same test: Marcel-lite MAE $\approx$ **0.0826**; train-mean MAE $\approx$ **0.0854**.

Count layer (projected-TBF exposure): expected-K MAE / RMSE / $R^2 \approx$ **1.790 / 2.213 / 0.168**; line Briers roughly **0.12–0.22** (lines 3.5–7.5).

Full-stack evaluation

Evaluation gate	Result
Estimator tuning	Keep baseline LightGBM defaults; Ridge α tuned and persisted
Walk-forward stack backtest	Mean expected-K MAE $\approx$ 1.778 (3 expanding 2024 windows; $\sigma \approx 0.036$)
Calibration	Mean ECE $\approx$ 0.024 (no recalibration)
Population policy	Metrics conditional on PA ≥ 9 (~3.5% of first pitchers excluded)

Takeaway

Leakage-safe Statcast $\rightarrow$ rate $\times$ exposure stack with nested chronological selection. Clears a Marcel-lite talent floor on chronological test; opponent lineup is the only leave-family-out family with both-fold within-fold bootstrap support. Absolute R^2 remains limited (~ 0.15). Portfolio claim: ML engineering under a hard pregame constraint.

Full write-up: [docs/paper/manuscript.pdf](#)